Problem Chosen	2025 MCM/ICM Summary Sheet	Team Control Number
B		2512884

Sustainable Tourism Model for Balanced Development

Summary

Juneau attracts a large number of tourists due to its rich tourism resources and convenient transportation. In 2023, it received 1.6 million cruise passengers, generating revenue of $375 million. However, this has also led to environmental and social problems, such as the retreat of glaciers, strain on infrastructure, and a decline in the quality of life of local residents. This research focuses on the sustainable tourism development of Juneau, Alaska, by constructing a mathematical model to balance economic benefits, resource conservation, and local welfare.

For Problem 1: The sustainable tourism model established in this study comprehensively **considers economic, environmental, and social factors**. It takes variables like the number of tourists and tourism revenue as **core elements** and describes the relationships among various factors through relevant formulas, including the model for the growth of the number of tourists, the calculation method of tourism revenue, and the expressions for environmental pressure, infrastructure pressure, and community satisfaction. **The Genetic Algorithm** is applied to solve the optimization problem. The objective function maximizes **tourism revenue** while taking into account **environmental protection, infrastructure carrying capacity, and social satisfaction**, and corresponding constraint conditions are set to ensure the feasibility of the model. The results show that the fitness of the population continuously improves during the iteration process of the genetic algorithm. The sensitivity analysis reveals that factors such as daily visitor limits, environmental protection willingness, and community development needs **have a significant impact** on the overall performance of the model.

For Problem 2: The model is applied to **Hainan, China**, and the parameters are adjusted according to the tourism characteristics of Hainan. Hainan attracts tourists with its unique beach scenery, warm climate, and rich culture. The application results indicate that the natural environment, historical culture, and social economic conditions of different regions will change the manifestation of the model, and the application of the model needs to **be adjusted and optimized according to local conditions**.

For Problem 3: Based on the model analysis, a reasonable daily tourist upper limit should be set: **the daily tourist limit between 650 - 700 people (234,000 - 252,000 people annually)** can, to a certain extent, achieve the growth of tourism revenue and simultaneously promote development in multiple aspects. The environmental protection coefficient should be increased, and investment in the protection of the natural environment should be enhanced. Taxes should be allocated reasonably for community project construction to balance community development needs, improve the quality of life of residents and community satisfaction, so as to **achieve the coordinated and sustainable development of tourism economy, environment, and society**.

Keywords: Managing Sustainable Tourism; Juneau; Genetic Algorithm; Multi-Objective Optimization Problem

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1 Introduction

1.1 Background

n academic fields of tourism and urban planning, sustainable tourism development in popular cities poses complex challenges. These cities must balance economic benefits with environmental protection, cultural heritage preservation, and resident welfare. Unregulated tourism growth may cause resource depletion, ecological degradation, and social unrest.

Figure 1 shows the main activity centers and points of interest in Juneau. This indicates the abundance of tourism resources in Juneau. Figure 2 shows he public transit routes and bus stops in Juneau. This illustrates the convenience of transportation connections between Juneau and other areas. Juneau has become a popular tourist destination due to its well-developed transportation and rich attractions. In 2023, it hosted 1.6 million cruise passengers, peaking at over 20,000 daily and generating $375 million in revenue. However, this surge has significantly impacted Juneau's environment and society, accelerating the retreat of the Mendenhall Glacier by eight football fields' length since 2007, straining local infrastructure, increasing carbon emissions, and deteriorating residents' quality of life due to housing shortages, higher prices, and overcrowding, leading to discontent and protests.



Figure 1: Activity Centers



Figure 2: Capital Transit Route and Bus Stop

Addressing sustainable tourism while conserving natural resources is an imperative for Juneau. This challenge is central to the objectives of our analytical model, highlighting the need for strategic interventions by local authorities and tourism boards.

1.2 Problem Restatement and Analysis

The significant impacts of overtourism on Juneau, Alaska, affect various aspects of the local environment and community. Through the analysis and investigation of the background and the intricate dynamics at play, the restated problem can be expressed as follows:

- Develop a mathematical model for sustainable tourism in Juneau to balance economic benefits, resource conservation, and local welfare.

  - Factor in the number of visitors, total revenue, and measures to stabilize the tourism industry, while also considering constraints and optimizing for sustainable outcomes.
  - Plan for the use of additional revenue, demonstrating how these funds can be reinvested to support conservation efforts, infrastructure improvements, and community development.
  - Perform a sensitivity analysis to identify the key factors affecting the sustainability of Juneau's tourism industry.

- Extend the model to enhance its applicability across a broader range of scenarios and to strengthen the model's explanatory power.
  - Refine the model to more comprehensively accommodate various tourist cities confronting the challenges of overtourism.
  - Adjust the model to promote less visited attractions, thereby achieving a better balance between economic, environmental, and social aspects.

- Compose a one-page memorandum to the Juneau Tourist Council.
  - Based on model analysis, predict future trends in Juneau's tourism industry, including aspects such as visitor numbers, tourism revenue, and environmental and social impacts.
  - Evaluate the impact of implemented or proposed measures on tourism sustainability, like hotel tax hikes, visitor fees, visitor caps, alcohol sales and consumption restrictions.
  - Provide advice on optimizing outcomes, balancing economic growth with environmental protection and resident welfare, and achieving sustainable tourism.

1.3 Overview of our work

The challenge before us involves devising a strategic management plan for sustainable tourism in Juneau, Alaska, amidst the complexities of environmental preservation and community welfare. The flow chart is show as figure 3.

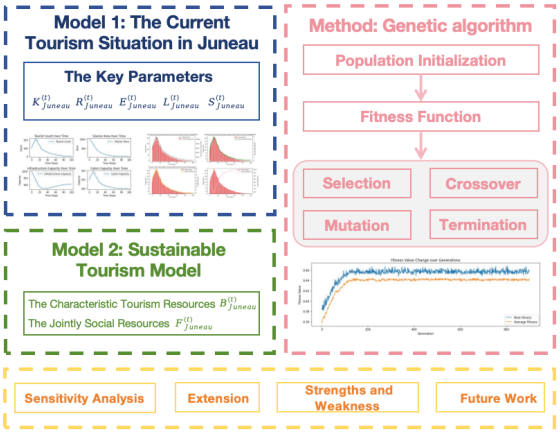


Figure 3: The Framework of Our Work

2 Assumptions

- **Assumption 1**: The changes in the number of tourists **have a sticky effect** on tourism revenue, the natural environment, social satisfaction and the level of infrastructure. Moreover, there exists a certain **linear relationship** between the number of tourists and tourism revenue, the natural environment, social satisfaction and the level of infrastructure.
- **Assumption 2**: Government policies, such as increased hotel taxes, visitor fees, caps on the number of daily visitors, and restrictions on alcohol sales and consumption, can **be effectively and rapidly implemented** and can be adjusted in a timely manner.
- **Assumption 3**: The local economy, society, and natural environment will not **experience significant changes** in the short term, such as economic crises, social unrest, or natural disasters.
- **Assumption 4**: The additional revenue from taxes can only be used to support conservation efforts, make improvements in infrastructure, and develop community programs, with no occurrences of corruption or waste.
- **Assumption 5**: Historical data can **be used to predict the trends** in future tourist numbers and tourism revenue. The parameters used in the model (such as tourists and carbon emission) have a certain degree of stability and predictability.

3 List of Notation

The important symbols used in this article and their meanings are shown in Table 1. There are some variables that are not listed here and will be discussed in detail in each section.

Table 1: Notation and Definitions

Symbol	Description
$K_i^{(t)}$	The number of tourists received by tourist attraction i at time t
$R_i^{(t)}$	The tourism revenue earned by tourist attraction i at time t
$T_i^{(t)}$	The tax revenue collected by tourist attraction i at time t
$c_i^{(t)}$	The consumption revenue obtained by tourist attraction i at time t
$E_i^{(t)}$	The level of environmental pressure of tourist attraction i at time t
$B_i^{(t)}$	The stock of characteristic resources of tourist attraction i at time t, such as glaciers
$P_i^{(t)}$	The carbon footprint generated by tourist attraction i at time t
$S_i^{(t)}$	The satisfaction level of local residents at tourist attraction i at time t
$F_i^{(t)}$	The level of infrastructure at tourist attraction i at time t
$\omega_{ij}^{(t)}$	The proportion of government expenditure allocated to department j by tourist attraction i at time t
$w_{ij}^{(t)}$	The weight coefficient of department j by tourist attraction i at time t

Notice: The index i refers to different tourist attraction. And the index $j = 1, 2, 3$ refers to the usage of additional revenue: environmental , infrastructure and community improvement.

4 The Current Tourism Situation in Juneau

The number of tourists is one of the core variables in the model. This paper defines V(t) as the number of tourists visiting Juneau each year. The increase in the number of tourists not only directly affects tourism revenue but also puts greater pressure on local infrastructure and may lead to more severe environmental burdens.

The number of tourists is usually influenced by a variety of factors, including but not limited to the attractiveness of the tourist destination (such as unique tourism resources like glaciers) and the convenience of transportation (whether the infrastructure is accessible). In the absence of restrictions, the growth of the number of tourists can be represented by an exponential model:

$$K_{Juneau}^{(t)} = K_{Juneau}^{(0)} e^{rt} \frac{B_{Juneau}^{(t)} + F_{Juneau}^{(t)}}{(B_{Juneau}^{(0)} + F_{Juneau}^{(0)}) \cdot tax} \tag{1}$$

Where $K^{(0)}$, $B^{(0)}$ and $F^{(0)}$ represents the initial level of tourists, glacier area and infrastructure. r denotes the growth rate of the number of tourists. And t represents time (measured in years).

Tourist expenditure is the primary source of tourism revenue for the City of Juneau. During their travels, tourists spend money on accommodations, dining, attraction tickets, transportation, shopping, and other aspects. These expenditures directly affect the city government's revenue. We assume that the consumption patterns of tourists are relatively stable, and therefore, the total revenue can be calculated based on the number of tourists and the average expenditure per tourist. Let the total revenue be denoted as R(t), which can be expressed as the product of the number of tourists and the average income per tourist:

$$R_{Juneau}^{(t)} = IK_{Juneau}^{(t)} \tag{2}$$

Where I represents the average expenditure of tourists in Juneau. Furthermore, the tourism revenue of Juneau can be obtained through means such as the collection of visitor fees and taxes in Juneau, which serves as additional tax revenue for the Juneau government. Another portion is allocated as income for residents. Specifically, this can be represented by the following formula:

$$\begin{aligned} T_{Juneau}^{(t)} &= tax \cdot R_{Juneau}^{(t)} \\ c_{Juneau}^{(t)} &= (1 - tax) \cdot R_{Juneau}^{(t)} \end{aligned} \tag{3}$$

Where tax represents the tourism tax rate established by the Juneau municipal government.

The impact of overtourism on the environment is a significant issue faced by the City of Juneau. Environmental stress is primarily manifested in aspects such as glacier melting and carbon emissions. The glacier area and carbon emissions can be expressed as the product of the number of tourists and the burden coefficient of each tourist on glacier melting and carbon emissions:

$$\begin{aligned} B_{Juneau}^{(t)} &= B_{Juneau}^{(t-1)} - mK_{Juneau}^{(t-1)} + \gamma_1 \cdot \omega_1 \cdot T_{Juneau}^{(t)}/10000 \\ P_{Juneau}^{(t)} &= P_{local}^{(t)} + pK_{Juneau}^{(t-1)} - \gamma_1 \cdot \omega_1 \cdot T_{Juneau}^{(t)}/10000 \end{aligned} \tag{4}$$

Where γ_1 represents the conversion rate of the Juneau government's expenditure in environment protection. The variables m and k represent the burden coefficients for each tourist's impact on

glaciers and carbon emissions, respectively. P_{local} represents the carbon emissions generated by the residents of Juneau.

Then the overall environmental pressure can be expressed as:

$$E_{Juneau}^{(t)} = (m+p)K_{Juneau}^{(t)} - \gamma_1 \cdot \omega_1 \cdot T_{Juneau}^{(t)}/10000 \tag{5}$$

With the increase in the number of tourists, the infrastructure of Juneau City will face increasing pressure. The infrastructure pressure mainly stems from the tourists' needs for accommodation, transportation, waste generation, and energy consumption. Let the infrastructure pressure be denoted as $L^{(t)}$, which can be represented as the product of the number of tourists and the burden coefficient per tourist on the infrastructure:

$$L_{Juneau}^{(t)} = \beta K_{Juneau}^{(t)} - \gamma_2 \cdot \omega_2 \cdot T_{Juneau}^{(t)}/10000 \tag{6}$$

Where γ_2 represents the conversion rate of the Juneau government's expenditure in infrastructure improvement. β represents the burden coefficient per tourist on the infrastructure. In an academic context, the level of infrastructure carrying capacity can be quantitatively described by:

$$F_{Juneau}^{(t)} = F_{Juneau}^{(t-1)} - nK_{Juneau}^{(t-1)} + \gamma_2 \cdot \omega_2 \cdot T_{Juneau}^{(t)}/10000 \tag{7}$$

Where n represents the burden coefficients for each tourist's impact on infrastructure.

Combining equations (1), (4), and (7), we can derive the temporal trajectories of tourist numbers, glacier area, infrastructure carrying capacity, and carbon emissions, as depicted in the figure 4. Initially, there is a sharp increase in tourist count and the carbon emissions. Following the peak, the tourist count and the carbon emissions gradually decrease, showing a downward trend towards the end of the time period. The glacier area and infrastructure capacity start at a high value and shows a significant decrease, indicating melting or reduction over time. The decline is steady and continues throughout the observed period.

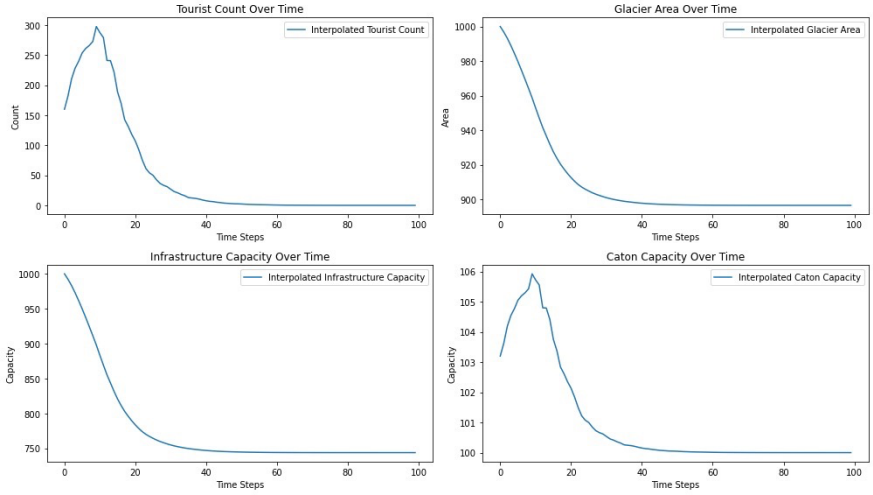


Figure 4: The Model Dynamic Result

With the increase in the number of tourists, social issues such as rising prices and increased housing costs may arise, which can put pressure on the quality of life of local residents. Therefore, we also need to consider community satisfaction, which is characterized by the burden on the socio-economic situation caused by each tourist and the number of tourists:

$$S_{Juneau}^{(t)} = 100 - sK_{Juneau}^{(t)} + \gamma_3 \cdot \omega_3 \cdot T_{Juneau}^{(t)}/10000 \tag{8}$$

Where γ_3 represents the conversion rate of the Juneau government's expenditure in community programs development. s represents the burden coefficients for each tourist's impact on community satisfaction.

Figure 5 respectively illustrates the relationships between tourist numbers and tourism revenue, environmental pressure, infrastructure pressure, and community satisfaction. Obviously, tourist numbers are positively correlated with tourism revenue, environmental pressure, and infrastructure pressure, while negatively correlated with community satisfaction.

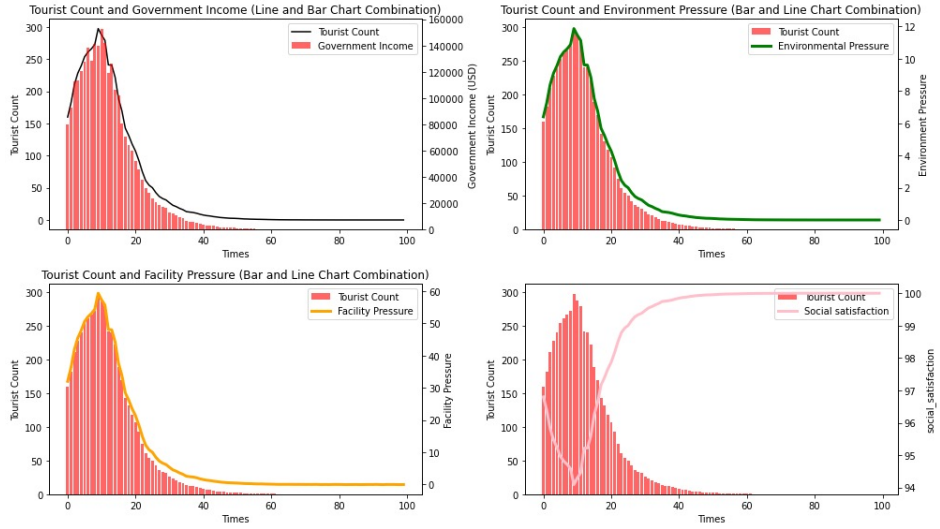


Figure 5: The Relationship between Tourist Numbers and Various Variables

4.1 Solution for the Tourism Situation

In the sustainable tourism model of Juneau, the core objective of the optimization problem is to maximize tourism revenue while ensuring that constraints such as environmental protection, infrastructure carrying capacity, and social satisfaction are met. This optimization problem is proposed to balance economic benefits and resource carrying capacity, so as to achieve the sustainable development of the tourism industry. Hence, the **OPTIMAL FUNCTION** is set as follows:

$$\max Z_{Juneau}^{(t)} = \frac{R_{Juneau}^{(t)} - R_{min}}{R_{max} - R_{min}} - w_1 \frac{E_{Juneau}^{(t)} - E_{min}}{E_{max} - E_{min}} - w_2 \frac{L_{Juneau}^{(t)} - L_{min}}{L_{max} - L_{min}} + w_3 \frac{S_{Juneau}^{(t)} - S_{min}}{S_{max} - S_{min}} \tag{9}$$

Where w_1, w_2 and w_3 represent the weight coefficient of environment conservation, infrastructure improvement and community programs development, respectively.

We not only need to consider how to increase revenue, but also need to balance environment, infrastructure, and society. To ensure the feasibility of the model and safeguard the sustainable development of Juneau, we introduces **CONSTRAINTS** on environmental protection $E_{constraint}$, infrastructure carrying capacity $L_{constraint}$, and socioeconomic $S_{constraint}$:

$$\begin{aligned} E_{Juneau}^{(t)} &\leq E_{constraint} \\ L_{Juneau}^{(t)} &\leq L_{constraint} \\ S_{Juneau}^{(t)} &\geq S_{constraint} \end{aligned} \tag{10}$$

Method: Genetic algorithm (GA) Genetic algorithm (GA) is a optimization technique inspired by the process of natural evolution. They work by simulating the selection, crossover, and mutation of potential solutions, known as individuals or chromosomes, over several generations. Each individual's fitness is evaluated based on a predefined objective function. The algorithm selects individuals with higher fitness to reproduce, combining their traits to produce offspring, while occasional mutations introduce diversity. The following is the pseudo-code (Algorithm 1) for this stage, which gives a good representation of the entire code flow.

Algorithm 1 *Sustainable Tourism System Model Based on Genetic Algorithm*

Input: Population size N, time steps $T = 1000$, crossover probability p_c, mutation probability p_m, growth rate range $[r_{min}, r_{max}]$

Initialization: Initialize population P with individuals having growth rate r

for $t = 1$ T **do**

  foreach *individual i in P* **do**

    Calculate indicators (number of tourists, income, environmental pressure, infrastructure pressure, social satisfaction) based on r_i Compute fitness $Fitness(i)$ according to growth rate - related indicators

  end

  Selection: Use roulette wheel selection. Select individuals based on $Fitness(i)$ to form P_{new}

  foreach *pair (i, j) in P_{new} with probability p_c* **do**

    Randomly pick mixing factor α Generate offspring k and l:

$$k = \alpha \cdot i + (1 - \alpha) \cdot j$$

$$l = (1 - \alpha) \cdot i + \alpha \cdot j$$

    Clip growth rates of k and l to $[r_{min}, r_{max}]$ Add k and l to P_{new}

  end

  foreach *individual i in P_{new} with probability p_m* **do**

    Sample ϵ from $\mathcal{N}(0, 0.01)$; Update $r_i = r_i + \epsilon$ and clip to $[r_{min}, r_{max}]$

  end

  $P \leftarrow P_{new}$

end

Output: Find individual with max $Fitness(i)$ as optimal plan; Predict long-term tourism trends based on it **Output:** Optimal tourism plan, long-term trend prediction

4.2 Results and Analysis

Perform numerical simulations 100 times, 500 times and 1000 times respectively. The optimal parameter settings and fitness values are shown in Table 2.

Table 2: Model output results

Times	K_{limit}	tax	ω_1	ω_2	ω_3	$Fitness(i)$
100	747.197	0.387	0.512	0.545	0.544	0.381
500	747.212	0.497	0.54	0.555	0.512	0.46
1000	747.236	0.497	0.533	0.566	0.505	0.458

Figure 6 shows the change in fitness values over generations after applying **the Genetic Algorithm** to the sustainable tourism model in Juneau. The figure contains two curves: the blue curve represents the best fitness in each generation, that is, the fitness value of the best - performing individual in each generation of the population; the orange curve represents the average fitness of each generation, that is, the average value of the fitness values of all individuals in each generation.

Obviously, as the number of Generation increases from 0 to 1000, both the best fitness and the average fitness generally show an upward trend. This indicates that as the genetic algorithm iterates, the fitness of the population is constantly improving, that is, the algorithm is constantly optimizing to find better solutions.

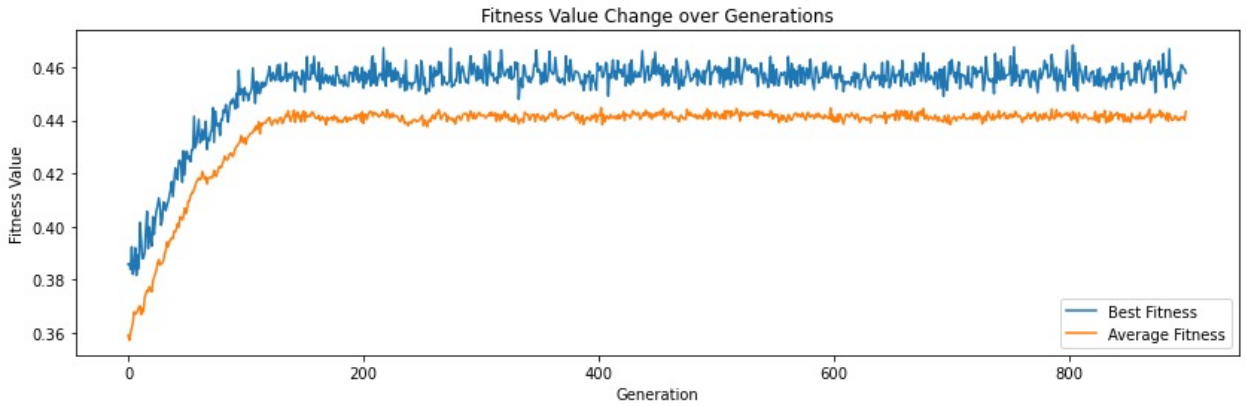


Figure 6: Result of the Genetic Algorithm

Figure 7 shows the genetic factors over generations. The figure consist of five sub-figures: Sub-figure 1 shows the changes in the upper limit of the number of tourists in each generation. Sub-figure 2 presents the changing trends of the tax rate in each generation. Sub-figure 3 shows the changes in the environment - related weight (ω_1) in each generation. Sub-figure 4 shows the changes in the infrastructure - related weight (ω_2) in each generation. Sub-figure 5 shows the changes in the community - related weight (ω_3) in each generation. In the sustainable tourism model of Juneau, different key factors are all dynamically adjusted to optimize the objective function.

Figure 8 shows the genetic factors distribution over generations. The figure consist of five sub-figures: Sub-figure 1 shows the distribution of the upper limit of the number of tourists in each generation. Sub-figure 2 shows the distribution of the tax rate in each generation. Sub-figure 3 shows the distribution of the environment - related weight (ω_1) in each generation. Sub-figure 4 shows the distribution of the infrastructure - related weight (ω_2) in each generation.. Sub-figure 5 shows the distribution of the community - related weight (ω_3) in each generation.

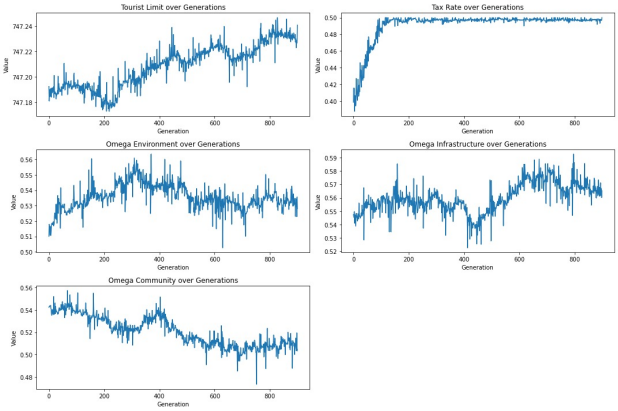


Figure 7: Genetic Factors Changes

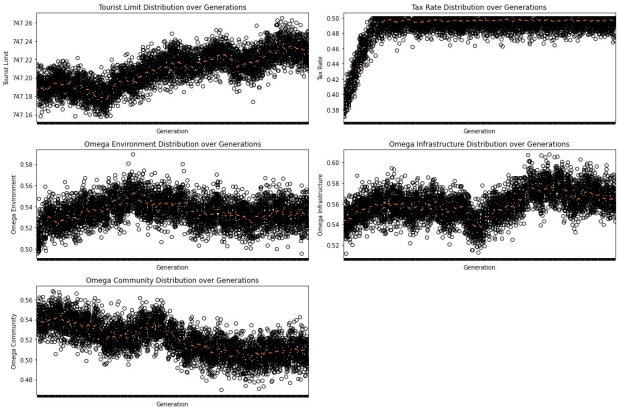


Figure 8: Genetic Factors Distribution

Figure 9 shows the population distribution in the last generation(heatmap). The horizontal axis in the figure represents the Tourist Limit, and the vertical axis represents the Tax Rate. The color - coding of the heatmap represents the Population Density. The colors range from black (representing a density of 0) to yellow (representing a relatively high density) and then to white (representing the highest density). The numerical values of the population density corresponding to different colors can be intuitively seen through the color scale bar on the right - hand side. From the figure, it can be observed that within specific combinations of the tourist limit and the tax rate, the population density is relatively high, indicating that the genetic algorithm has a greater tendency to generate individuals under these parameter combinations during the iteration process. For example, when $tax \in [0.496, 0.499]$ and $K_{limit} \in [747.225, 747.227]$, the colors are brighter, suggesting that these parameter combinations are more common in the last generation.

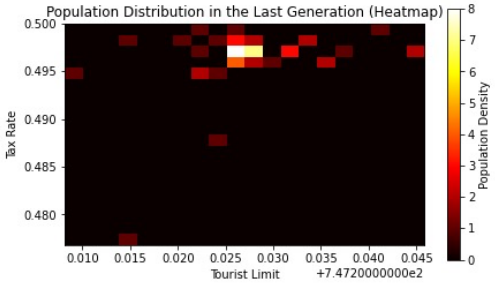


Figure 9: The Population Distribution in the Last Generation(Heatmap)

5 Sustainable Tourism Model: Economic, Environment and Society

In the discussion of which factors are most important for sustainable tourism in Juneau, the number of tourists takes center stage. It directly determines tourism revenue, with more tourists typically resulting in higher income. However, it also has far-reaching negative impacts. It burdens local infrastructure, causing issues like overcrowding in transportation and accommodation. Environmentally, it speeds up glacier melting and raises carbon emissions. Socially, it can reduce community satisfaction due to problems such as increased housing costs and overcrowding. Since it influences economic, environmental, and social aspects so comprehensively, effectively managing the number of tourists is crucial for achieving sustainable tourism in Juneau, outweighing other factors in terms of its overall impact. Now, we abstract the model to make it adaptable to a wider range of tourist attractions as a **Sustainable Tourism Model**.

The growth of the number of tourists in tourist attraction i can be represented by equation 11:

$$K_i^{(t)} = K_i^{(0)} e^{rt} \frac{B_i^{(t)} + F_i^{(t)}}{(B_i^{(0)} + F_i^{(0)}) \cdot Measures_i} \tag{11}$$

Where B_i represents the characteristic tourism resources of tourist attraction i, F_i represents the social resources that are jointly used by local residents and tourists at tourist attraction i, and $Measure_i$ represents the measures implemented by tourist attraction i to achieve sustainable tourism development.

Similarly, the changes in characteristic tourism resources and the social resources jointly used by local residents and tourists can be represented by the equation 12:

$$\begin{aligned} B_i^{(t)} &= B_i^{(t-1)} - mK_i^{(t-1)} + \gamma_1 \cdot \omega_1 \cdot Useage(Measure_i^{(t)}) \\ F_i^{(t)} &= F_i^{(t-1)} - nK_i^{(t-1)} + \gamma_2 \cdot \omega_2 \cdot Useage(Measure_i^{(t)}) \end{aligned} \tag{12}$$

Where $Useage(Measure_i^{(t)})$ represents the protective effect of government measures on tourism resources and social resources.

The income, natural environmental pressure, infrastructure pressure, and community satisfaction included in the model can be transferred to this model, that is equation 13:

$$\begin{aligned} R_i^{(t)} &= IK_i^{(t)} \\ E_i^{(t)} &= (m+p)K_i^{(t)} - \gamma_1 \cdot \omega_1 \cdot Useage(Measure_i^{(t)}) \\ L_i^{(t)} &= \beta K_i^{(t)} - \gamma_2 \cdot \omega_2 \cdot Useage(Measure_i^{(t)}) \\ S_i^{(t)} &= 100 - sK_i^{(t)} + \gamma_3 \cdot \omega_3 \cdot Useage(Measure_i^{(t)}) \end{aligned} \tag{13}$$

Thus, we can still establish the objective function that needs to be maximized and the constraints, namely equation 14:

$$\begin{aligned} \text{maximize} \quad & Z_i^{(t)} = Function(R_i^{(t)}, E_i^{(t)}, L_i^{(t)}, S_i^{(t)}) \\ \text{subject to} \quad & E_i^{(t)} \le E_{constranint}, L_i^{(t)} \le L_{constranint}, S_i^{(t)} \ge S_{constranint} \end{aligned} \tag{14}$$

6 Sensitivity Analysis: Dealing with More Complex Situations

In the pursuit of sustainable tourism development, a multitude of factors interact in complex ways. Sensitivity analysis serves as a crucial tool to understand how changes in key variables impact the overall model performance. By examining different aspects such as daily visitor limits, environmental protection willingness, and community development needs, we can gain insights into the delicate balance required for a sustainable tourism ecosystem.

6.1 The Daily Visitor Limits Measures

The figure shows the results of the sensitivity analysis for the visitor number limit (K_{limit}). The horizontal axis represents the daily visitor limit. The vertical axis represents the best fitness value. As the daily visitor limit gradually increases from 600, the best fitness value generally shows an upward trend. When the daily visitor limit is around 600-650, the best fitness value fluctuates between 0.30 and 0.35; when the limit increases to the range of 650-750, the best fitness value fluctuates between 0.35 and 0.45 with a more obvious upward trend; between 750 and 800, the best fitness value fluctuates between 0.40 and 0.45.

This indicates that the magnitude of the visitor - number limit plays an important role in local sustainable development. Its transmission path is to attract more visitors to consume at tourist attractions, achieve the growth of tourism revenue, and thus realize the growth of government tax revenue and distribution, thereby promoting environmental protection, infrastructure construction, and community development.

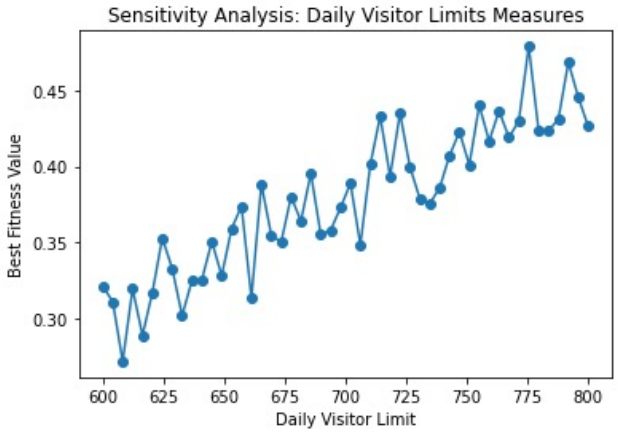


Figure 10: Sensitivity Analysis: Daily Visitor Limits Measures

6.2 Growth or Shrinkage in Environmental Protection Willingness

Figure 11 shows the sensitivity - analysis results for the environmental - protection coefficient (γ_1). The horizontal axis represents the environmental - protection coefficient (γ_1), and the vertical axis represents the best fitness value. As the coefficient increases from 0.400, the best fitness value generally rises. In the 0.400-0.450 range, it fluctuates between 0.41 and 0.43; in the 0.450-0.550 range, it fluctuates between 0.43 and 0.45 with a more obvious upward trend; in the 0.550-0.600 range, it fluctuates between 0.45 and 0.47 and reaches a relatively high value near 0.600.

Strengthening the influence of factors related to the protection of the natural environment has a positive effect on improving the overall performance of the model. Maintaining the degree of environmental protection within an appropriate range helps to ensure sustainable development.

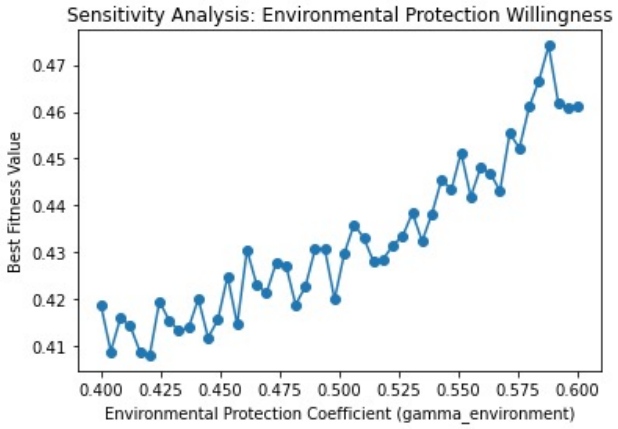


Figure 11: Sensitivity Analysis:Environmental Protection Willingness

6.3 Increase or Decrease of Community Development Needs

The figure 12 shows the results of the sensitivity analysis for community development needs (γ_3). The horizontal axis of the chart represents the Community Development Coefficient (γ_3), and the vertical axis represents the Best Fitness Value. As the community development coefficient gradually increases starting from 0.400, the best fitness value fluctuates within the range of 0.42-0.44, but remains relatively stable overall without significant changes.

Therefore, maintaining the community development coefficient (γ_3) within an appropriate range can effectively safeguard the needs of community residents, which is of positive significance for ensuring sustainable development and avoiding possible adverse changes in aspects such as model fitness caused by excessive neglect of the needs of community residents.

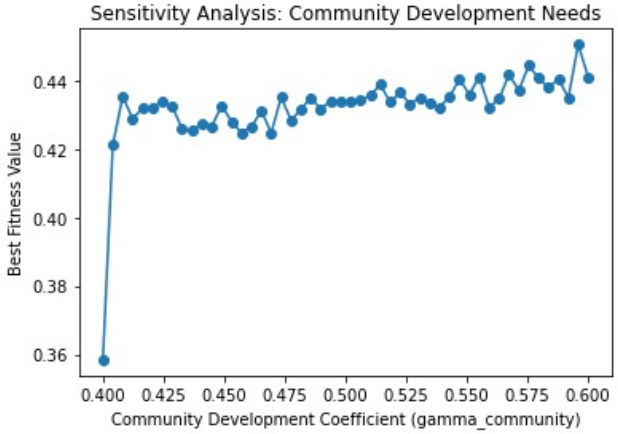


Figure 12: Sensitivity Analysis: Community Development Needs

7 Extension: Enhancing the Breadth of Model Application

In order to enhance the universality of the model, this paper modifies the Sustainable Tourism Model provided in Section 5 and applies it to Hainan in China and attractions with a relatively small number of visitors. **The 2024 Hainan Tourism Market Analysis Report** shows that in the first three quarters of 2024, the tourism popularity in Hainan, China, continued to rise. Among them, the ticket booking popularity of key tourist attractions in the province increased by nearly 60% year-on-year, and the booking popularity of hotels and homestays in places like Haikou, Sanya, and Wanning also witnessed a significant increase.

7.1 Applying the Model to Hainan, China

Figure 13 shows the geographical location, tourist attractions and transportation facilities in Hainan, China. With its unique beach scenery, warm climate and cultural history, Hainan has become one of the popular tourist destinations for Chinese people in winter. According to statistics, during the New Year's Day holiday in 2024 alone, Haikou City, Hainan Province, China, received 461,900 tourists, a year-on-year increase of 109.3%; and the total tourism revenue reached 544 million yuan, a year-on-year increase of 92.1%.



Figure 13: China Hainan Tourist Map

According to the tourism characteristics of Hainan, China, we define its characteristic tourism resources as **Beach Scenery and Cultural Relics**, and the social resources jointly used by tourists and local residents as **Transportation Facilities**. Obviously, the more tourists there are, the lower the quality of beach scenery and cultural relics that each person can enjoy, and the greater the carrying pressure on transportation facilities. Assume that tax is still applicable in Hainan to maintain the sustainable development of tourist attractions.

Hence, The number of tourists in Hainan is determined by the local tourist growth rate, the quality of beach scenery, the preservation of cultural relics, and the carrying capacity of transportation facilities, as shown in equation 15.

$$K_{Hainan}^{(t)} = K_{Hainan}^{(0)} e^{rt} \frac{Beach_{Hainan}^{(t)} + Relics_{Hainan}^{(t)} + Trans_{Hainan}^{(t)}}{(Beach_{Hainan}^{(0)} + Relics_{Hainan}^{(0)} + Trans_{Hainan}^{(0)}) \cdot tax} \tag{15}$$

Further, the relationships among the beach scenery, cultural relics in Hainan, and the carrying capacity on transportation facilities and the number of tourists are shown in Equation 16.

$$\begin{aligned} Beach_{Hainan}^{(t)} &= Beach_{Hainan}^{(t-1)} - mK_{Hainan}^{(t-1)} + \gamma_1 \cdot \omega_1 \cdot T_{Hainan}^{(t)}/10000 \\ Relics_{Hainan}^{(t)} &= Relics_{Hainan}^{(t-1)} - vK_{Hainan}^{(t-1)} + \gamma_2 \cdot \omega_2 \cdot T_{Hainan}^{(t)}/10000 \\ Trans_{Hainan}^{(t)} &= Trans_{Hainan}^{(t-1)} - nK_{Hainan}^{(t-1)} + \gamma_2 \cdot \omega_2 \cdot T_{Hainan}^{(t)}/10000 \end{aligned} \tag{16}$$

It should be noted that since cultural relics are part of the cultural landscape, they are not included in the consideration of the natural environment. Instead, they are incorporated into the consideration of infrastructure. Therefore, its tax revenue distribution coefficient and weight coefficient are ω_2 and γ_2 respectively.

The tourism revenue, environmental pressure, infrastructure pressure and community satisfaction can be expressed by equation 17.

$$\begin{aligned} R_{Hainan}^{(t)} &= IK_{Hainan}^{(t)} \\ E_{Hainan}^{(t)} &= mK_{Hainan}^{(t)} - \gamma_1 \cdot \omega_1 \cdot T_{Hainan}^{(t)}/10000 \\ L_{Hainan}^{(t)} &= (n+v)K_{Hainan}^{(t)} - \gamma_2 \cdot \omega_2 \cdot T_{Hainan}^{(t)}/10000 \\ S_{Hainan}^{(t)} &= 100 - sK_{Hainan}^{(t)} + \gamma_3 \cdot \omega_3 \cdot T_{Hainan}^{(t)}/10000 \end{aligned} \tag{17}$$

To facilitate the comparison of the contributions of the four parameters, we standardize the four parameters in equation 17 using the maximum-minimum normalization method. The normalization formula is shown in equation 18.

$$X_{normal}^{(t)} = \frac{X_{Hainan}^{(t)} - X_{min}}{X_{max} - X_{min}} \tag{18}$$

The optimal objective function and constraints of the sustainable tourism development model in Hainan are shown in equation 19.

$$\begin{aligned} \text{maximize} \quad & Z_{Hainan}^{(t)} = R_{normal}^{(t)} - w_1 E_{normal}^{(t)} - w_2 L_{normal}^{(t)} + w_3 S_{normal}^{(t)} \\ \text{subject to} \quad & E_{Hainan}^{(t)} \leq E_{constranint}, L_{Hainan}^{(t)} \leq L_{constranint}, S_{Hainan}^{(t)} \geq S_{constranint} \end{aligned} \tag{19}$$

Perform numerical simulations 100 times, 500 times and 1000 times respectively. The optimal parameter settings and fitness values are shown in Table 3.

Table 3: Model output results

Times	K_{limit}	tax	ω_1	ω_2	ω_3	$Fitness(i)$
100	681.868	0.497	0.487	0.554	0.386	0.524
500	672.713	0.497	0.393	0.642	0.532	0.904
1000	701.005	0.497	0.562	0.566	0.495	0.953

Figure 14 shows the result of the Genetic Algorithm in Hainan. The blue line represents Best Fitness, and the orange line represents Average Fitness. As can be seen from the graph, in the initial stage, there are large fluctuations in the best fitness value and the average fitness. However, as the number of generations increases, they gradually tend to stabilize. Overall, the best fitness is higher than the average fitness. This indicates that when the Sustainable Tourism Model is applied to Hainan, it still has good convergence properties.

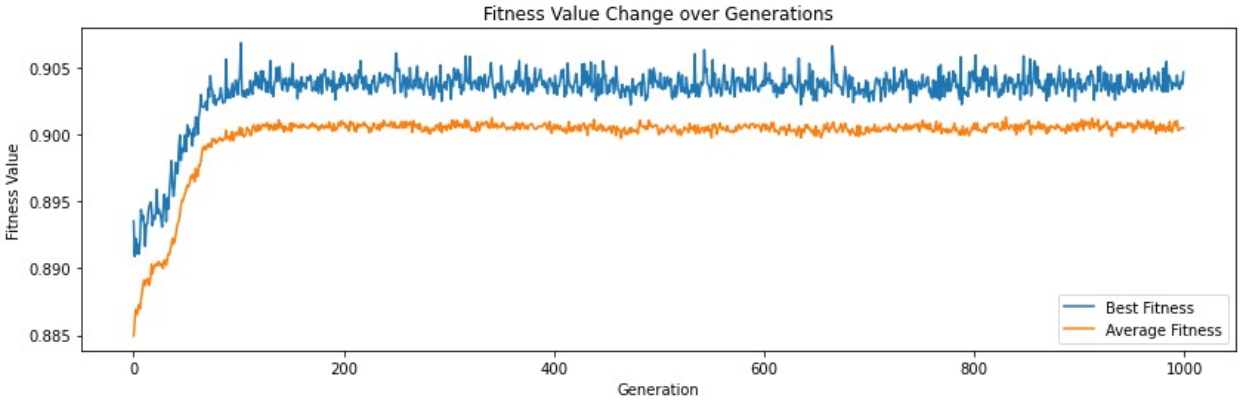


Figure 14: Result of the Genetic Algorithm in Hainan

When comparing the application of the Sustainable Tourism Model in Juneau and Hainan, the choice of location significantly impacts the model results in terms of the natural environment, historical culture, and social economy.

In terms of the natural environment, Juneau has glaciers ecosystem. When developing tourism activities such as glacier hiking , it is necessary to protect the glaciers. Hainan has tropical beaches and abundant marine life. The key to applying the model lies in protecting the sand quality and water quality of the beaches.

Regarding historical culture, the indigenous culture and gold-rush culture in Juneau are important tourism resources. Hainan has a long history and diverse cultures, including ethnic minority cultures such as those of the Li and Miao ethnic groups, and the Maritime Silk Road culture. When applying the model, attention should be paid to their protection and inheritance, and cultural exhibition centers can be set up.

Hence, the natural resources and historical cultures of different locations will change the manifestation of the model. Its application needs to be adjusted and optimized according to local conditions to achieve economic development, environmental protection, and cultural inheritance, and improve the quality of life of local residents.

7.2 Promote Less-Visited Attractions and Regions for Balanced Development

Figure 15 shows the promotion model we developed for tourist attractions with fewer visitors.

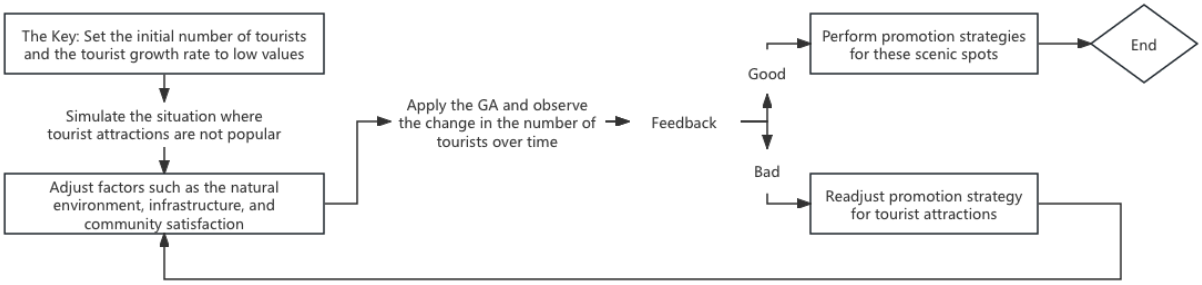


Figure 15: Flowchart for Promoting Tourist Attractions with Fewer Visitors

If we aim to utilize the model to promote attractions or regions with fewer tourists for a more balanced development, we can begin by setting the initial number of tourists and the tourist growth rate at relatively low values. Setting the initial number of tourists and the growth rate low is essentially simulating a situation where the tourist attractions are not popular among tourists. After setting these parameters, we can then introduce factors such as the natural environment, infrastructure, and community satisfaction into the model. When adjusting these factors within the model, we can observe the gradual change in the number of tourists over time. If the model shows a positive response, that is, the number of tourists steadily increases while maintaining a sustainable level in terms of environmental and social impacts, then we can implement these strategies at real - world destinations. In this way, we can promote attractions with fewer tourists and achieve a more balanced development of tourism distribution.

8 Discussion

8.1 Strengths and weaknesses

8.1.1 Strengths

- **Scientific and Rational Model Construction:** A mathematical model considering economic, environmental, and social factors is built for Juneau's sustainable tourism. The objective function maximizes tourism revenue while accounting for other aspects, with clear constraints. The genetic algorithm is used, offering a sound quantitative framework for sustainable tourism development.

- **Reliable and Diverse Data Visualization:** Various types of charts are used to visually present variable relationships, model dynamics, sensitivity analysis results, and differences in model

applications across different regions, improving the efficiency of information dissemination and enhancing readability and persuasiveness.

- **Comprehensive and Multifaceted Analysis:** The report analyzes Juneau's tourism in detail, covering aspects like tourist numbers, revenue, environmental impact, infrastructure pressure, and community satisfaction, using formulas and charts. Sensitivity analysis studies factor impacts for policy making. Applying the model to Hainan and comparing differences boosts its universality and practicality.

8.1.2 Weakness

- **Overly Idealized Model Setting:** Our models are intended to capture the essence of the problems through idealized and abstract descriptions first. Therefore, the model is relatively complete in theory, but there maybe a deviation from the actual, for the model lacks practical testing.

8.2 Future Work

In future work, emphasis will be placed on deeply improving the model. On the one hand, more complex real world factors such as seasonal changes, global economic fluctuations, and public health emergencies will be comprehensively incorporated. We will consider how these factors dynamically affect the number of tourists, tourism revenue, and environmental conditions in Juneau and other tourist destinations. At the same time, cutting edge algorithms like machine learning will be used to replace or optimize existing algorithms, improving the accuracy and reliability of the model's predictions. On the other hand, multi source data such as social media data and geographic information data will be widely integrated. From a more comprehensive perspective, we will capture the changing trends of the tourism market and the behavioral characteristics of tourists, enabling the model to more realistically and meticulously reflect the actual situation. In addition, the optimized model will be used to conduct long term evaluations of various potential policies. This includes the combined effects of different tax policies, tourist management strategies, and environmental protection measures, providing more forward looking and practical decision making references for policymakers.

We also need to expand the content of the report. On the one hand, we will focus on the coordinated development of regional tourism, deeply study the cooperation model between Juneau and its surrounding areas, and achieve the overall upgrade and sustainable development of the tourism industry by integrating tourism resources and optimizing tourism routes. We will construct a regional tourism cooperation model, analyze in detail the economic, environmental, and social impacts of different cooperation strategies on various tourist destinations in the region, explore a win - win cooperation mechanism, and summarize successful experiences and existing problems in combination with actual cases to provide references for other tourist destinations. On the other hand, we will deeply explore the impact of tourist behavior and tourism experience on sustainable tourism. By comprehensively using methods such as questionnaires, interviews, and big data analysis, we will accurately understand tourists' consumption behaviors, preferences, and satisfaction levels, providing a strong basis for tourism product design and service quality improvement.

9 Conclusion

The sustainable tourism model constructed in this study comprehensively considers economic, environmental, and social factors. The genetic algorithm is used to solve the optimization problem, and results under different parameter settings are obtained through multiple groups of numerical simulations. Judging from the model results, this model can better reflect the relationships among various factors and provides a quantitative framework for sustainable tourism development at the theoretical level. For example, the sensitivity analysis clearly demonstrates the impacts of factors such as daily visitor limits, environmental protection willingness, and community development needs on the overall performance of the model, providing a strong basis for policy - making. At the same time, when the model is applied to Hainan, it is found through comparison that the natural environment, historical culture, and social - economic conditions of different regions will change the manifestation of the model, indicating that the model has a certain degree of universality, but it needs to be adjusted and optimized according to local conditions when applied.

Regarding the sustainable tourism development of Juneau, the following suggestions are put forward based on the results of model analysis: In terms of tourist number control, a reasonable daily tourist upper limit can be set according to the results of the sensitivity analysis. For example, when the daily tourist limit is set between 650-700 people(yearly 234,000-252,000), it can, to a certain extent, achieve the growth of tourism revenue and simultaneously promote environmental protection, infrastructure construction, and community development. In terms of environmental protection, increase the environmental protection coefficient, increase investment in the protection of the natural environment, maintain the stability of the glacier ecosystem, and reduce the impact of tourism activities on glacier melting and carbon emissions. In terms of community development, allocate taxes reasonably for community project construction, balance community development needs, guarantee the quality of life of residents, and improve community satisfaction. Through the comprehensive implementation of these measures, it is expected to achieve the coordinated and sustainable development of Juneau's tourism economy, environment, and society.

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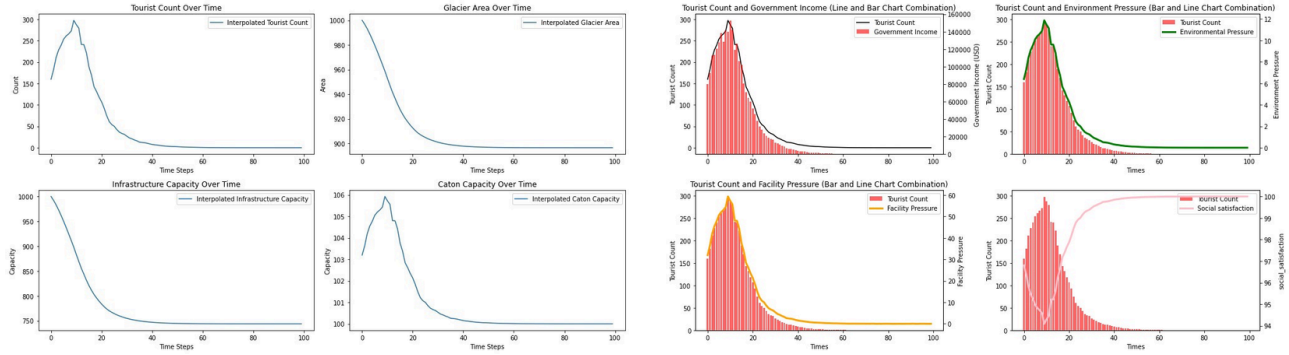
Memorandum to Juneau, AK

TO: Juneau Tourist Council
FROM: MCM Team #2512884
DATE: January 27, 2025
SUBJECT: Forecasting and Optimization Proposal for Sustainable Tourism Development in Juneau

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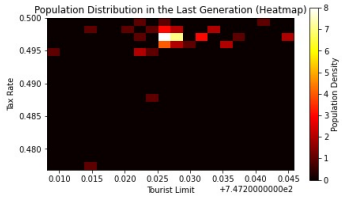
Current Situation and Prediction

1.67 million visitors in 2023, contributing to economic growth but increasing environmental pressure, especially on Mendenhall Glacier. Overloaded infrastructure and waste management issues affect the local residents' quality of life. According to the Figure 5, if the government doesn't take any measures, **the number of tourists will reach its peak in 11 years and then enter an irreversible downward trend**, which means that the sustainable development of Juneau's tourism industry can't be achieved.



The Effects of Various Measures

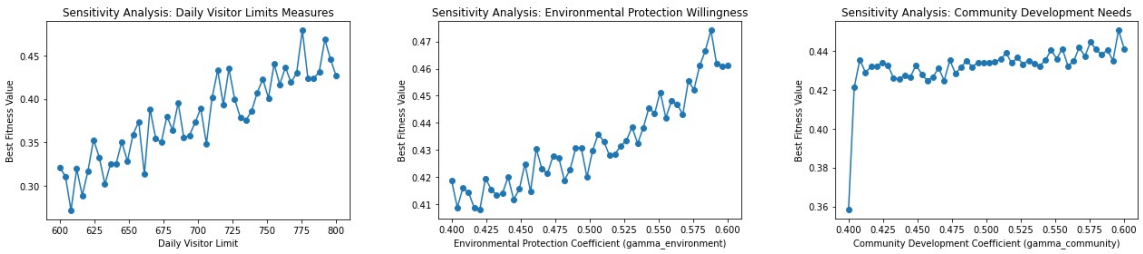
- **Visitor Caps:** Setting a daily visitor limit can have a positive impact. When the daily tourist limit is between 650-700 people (234,000-252,000 people annually), it can achieve a balance. It allows for the growth of tourism revenue while promoting development in environmental protection, infrastructure construction, and community development.



- **Tax Measures:** Hotel tax hikes and visitor fees can generate additional revenue. Generally, the tax rate is between 0.496-0.497. If allocated properly, this revenue can be used for conservation efforts, infrastructure improvements, and community development projects, which can enhance the overall sustainability of tourism. However, if the taxes are too high, they may deter some tourists.

Optimization Recommendation

Based on the sensitivity analysis, set a reasonable daily tourist upper limit. This limit should be adjusted periodically according to the carrying capacity of the environment, infrastructure, and community. Increase the environmental protection coefficient and invest more in natural environment protection. This can help maintain the stability of the glacier ecosystem and reduce the negative impacts of tourism on the environment. Allocate taxes reasonably for community project construction. By doing so, the needs of the community can be balanced, the quality of life of residents can be guaranteed, and community satisfaction can be improved.



Report on Use of AI

The team #2512884 used ChatGPT (Nov 5, 2023 version, ChatGPT-4) to assist in completing the following three tasks. First, it generated structured initial ideas for Problem B. Second, it provided channels or websites for collecting tourism data in Juneau. Third, it inquired about the basic logic of the genetic algorithm. The specific ways of Query and Output are as follows.

OpenAI ChatGPT (Nov 5, 2023 version, ChatGPT-4)

Query1 <What's your opinion in building a model for a sustainable tourism industry in Juneau, Alaska?>

Output <Building a model for sustainable tourism in Juneau is a fantastic and timely project, especially given the increasing emphasis on balancing economic growth with environmental and social well-being. Juneau's unique location, with its natural beauty and reliance on tourism, makes it an ideal case study. Here are a few thoughts on key components to consider when modeling for sustainability:

- **Economic Impact**: The tourism sector can drive significant income for the local economy, but it's essential to measure how this income is distributed. Ensuring that local businesses and residents benefit from tourism, rather than large, external companies, could help in making tourism more sustainable. You might want to model how tourism affects wages, employment rates, and the revenue streams for small businesses.
- **Environmental Sustainability**: Given Juneau's location in an ecologically sensitive area, preserving its natural environment is crucial. You could model the environmental impact of different tourism levels, like carbon footprints, waste management, and the pressure on local wildlife. You could also look at strategies like eco-tourism or regenerative tourism, where the goal is to leave the place better than it was found.
- **Social and Cultural Impacts**: How does tourism affect the local community, both positively and negatively? Are there any challenges like overcrowding, strain on public services, or loss of local culture? A key part of your model might focus on minimizing these negative social impacts. For example, promoting responsible tourism behaviors or establishing cultural preservation initiatives can help mitigate these issues.
- **Optimization**: You're working on an optimization problem, so finding the sweet spot between these factors—income, environmental impact, and social good—will be key. You could consider creating an objective function that balances these competing interests, adjusting for things like peak season vs off-season demand, types of tourists, or infrastructure capacity.
- **Long-term Sustainability**: Consider including a time element in your model, as tourism sustainability is not just about the present, but the future. What strategies can help ensure that tourism in Juneau remains viable for decades to come? Things like conservation efforts, investments in renewable energy, and diversifying the economy beyond tourism could play a role.>

Query2 <Where can tourism data for Juneau be found?>

Output <Tourism data for Juneau can be found from the following sources:

- City Government and Local Tourism Bureau: Juneau's official website or the Alaska Tourism Office may provide relevant tourism statistics.
- Tourism Research Reports and Surveys: Research institutions or academic organizations may publish studies on tourism in Juneau or Alaska.
- Local Tourism Businesses and Industry Associations: Hotels, tour operators, or cruise companies often collect data on visitor traffic, spending, and demographics.
- National Park Service (NPS): If Juneau has national parks or protected areas, the NPS may offer data on visitor numbers and activities.

- Online Travel Platforms and Social Media: Websites like TripAdvisor, Airbnb, and Instagram, while not entirely official, provide valuable information on visitor activities, accommodations, and reviews.>

Query3 <What is the basic logic of a genetic algorithm(GA)?>

Output <The basic logic of a genetic algorithm (GA) follows the principles of natural selection and evolution. Here's a breakdown of its core steps:

- **Initialization**: A population of candidate solutions (individuals) is created, usually randomly. Each individual is represented by a chromosome, typically encoded as a string (binary, real numbers, etc.).
- **Selection**: Individuals are selected based on their fitness, which measures how well they solve the problem at hand. The higher the fitness, the higher the probability that an individual will be selected for reproduction. Common selection methods include roulette wheel selection, tournament selection, and rank-based selection.
- **Crossover (Recombination)**: Pairs of individuals (parents) are chosen to produce offspring. The crossover operator combines parts of two parent chromosomes to create new offspring. This simulates genetic recombination, which introduces new solutions by blending the parents' traits.
- **Mutation**: After crossover, a small random change (mutation) is applied to some of the offspring. This introduces genetic diversity into the population and helps avoid premature convergence to suboptimal solutions.
- **Replacement**: The offspring (new generation) replace some or all of the old generation. The process may involve keeping the best individuals (elitism) or replacing the entire population.
- **Termination**: The algorithm repeats the selection, crossover, mutation, and replacement steps for a number of generations or until a termination condition is met, such as reaching a satisfactory solution or a maximum number of generations.

This iterative process leads to the gradual evolution of better solutions to the problem.>